Exercise: Designing a Database for an Online Movie Rental Platform

You have been hired to design a database for an online movie rental platform, similar to Amazon Prime Video. This platform allows users to rent movies that belong to different categories. It manages a large inventory of movies, and supports multiple users, payments, and rentals. Movies should be organized by category, genre, language and actors. The database should also include data about the contact information of users such as emails, phone numbers, addresses, etc.

<u>Part 1: Gathering Requirements</u>

Gather all business and design requirements needed to design the DB

Task:

- What are the primary operations that the platform should support?
- Who are the key relations/tables?
- What core data/attributes must be captured by the system?
- What cardinality ratios and participation constraints must be considered?

<u>Part 2: Building the Relational Data Model</u>

Using the gathered requirements, design a relational data model that captures all necessary information.

Task:

- Identify the major entities (e.g., Customer, Movie, Rental, ….).
- Identify key attributes for each entity (e.g., Movie: Title, Release Year, Genre, …).
- Determine the cardinality (e.g., one-to-many, many-to-many) for each relationship.
- Determine the participations requirements between relations/tables.
- Define primary and foreign keys for each table.
- Design the schema of the database, follow this template $r(a_1, a_2, \ldots, a_n)$

<u>Part 3: Design the ER Diagram</u>

Using the entities and relationships defined in the previous step, create an ER diagram to visually represent the database structure.

Task: Use draw.io to construct the Crow’s foot diagram.